Information and Software Technology 173 (2024) 107492

Contents lists available at ScienceDirect

Information and Software Technology

journal homepage: www.elsevier.com/locate/infsof

Understanding the landscape of software modelling assistants for MDSE tools: A systematic mapping

David Mosquera [a,*], Marcela Ruiz [a], Oscar Pastor [b], Jürgen Spielberger [a]

[a] ZHAW Zurich University of Applied Sciences, Gertrudstrasse 15, Winterthur 8400, Switzerland
[b] PROS-VRAIN: Valencian Research Institute for Artificial Intelligence, Universitat Politècnica de València, Camí de Vera, s/n, València 46022, Spain

ARTICLE INFO

Keywords:
Modelling assistance
Model-driven development
Systematic mapping
State of the practice
Low code
No-code

ABSTRACT

Context: Model Driven Software Engineering (MDSE) and low-code/no-code software development tools promise to increase quality and productivity by modelling instead of coding software. One of the major advantages of modelling software is the increased possibility of involving diverse stakeholders since it removes the barrier of being IT experts to actively participate in software production processes. From an academic and industry point of view, the main question remains: What has been proposed to assist humans in software modelling tasks?
Objective: In this paper, we systematically elucidate the state of the art in assistants for software modelling and their use in MDSE and low-code/no-code tools.
Method: We conducted a systematic mapping to review the state of the art and answer the following research questions: i) how is software modelling assisted? ii) what goals and limitations do existing modelling assistance proposals report? iii) which evaluation metrics and target users do existing modelling assistance proposals consider? For this purpose, we selected 58 proposals from 3.176 screened records and reviewed 17 MDSE and low-code/no-code tools from main market players published by the Gartner Magic Quadrant.
Result: We clustered existing proposals regarding their modelling assistance strategies, goals, limitations, evaluation metrics, and target users, both in research and practice.
Conclusions: We found that both academic and industry proposals recognise the value of assisting software modelling. However, documentation about MDSE assistants' limitations, evaluation metrics, and target users is scarce or non-existent. With the advent of artificial intelligence, we expect more assistants for MDSE and low-code/no-code software development will emerge, making imperative the need for well-founded frameworks for designing modelling assistants focused on addressing target users' needs and advancing the state of the art.

1. Introduction

Model Driven Software Engineering (MDSE) and low-code/no-code tools are software development tools that use models as the main input to generate software. That includes model-driven development tools, model-based development tools, low-code tools, and no-code tools, among others. MDSE tools aim to increase software development teams' productivity and decrease software time-to-market [1]. However, MDSE tools still require significant effort from software development teams in creating, maintaining, and debugging models. This is why modelling assistance becomes crucial, aiming to alleviate the modelling effort and boost the adoption of MDSE tools for software development. This mirrors Integrated Development Environments (IDEs) for coding, where features like code autocompletion, syntax highlighting, and debugging tools are essential components. Below we provide the definition of modelling assistance used in this paper:

Modelling assistance is the strategy—i.e., any method, technique, framework, or guideline—that aims to assist humans during software modelling tasks in MDSE tools.

Nevertheless, MDSE tools are not yet widely adopted in practice. This lack of adoption motivated different empirical research efforts to discover the benefits of such tools. Results [2,3] show that MDSE tools

Abbreviations: MDSE: model driven software engineering; IDE: integrated development environments; MRQ: main research question; GMQ: gartner magic quadrant; RQ: research question; I: inclusion criteria; E: exclusion criteria; G: goal; L: limitation; NS: not specified; MDE: model driven engineering; M: evaluation metric; U: target user; NE: not evaluated; LE: leaders; C: challengers; V: visionaries; NP: niche players; NF: not found

* Corresponding author.
E-mail address: mosq@zhaw.ch (D. Mosquera).

https://doi.org/10.1016/j.infsof.2024.107492
Received 11 December 2023; Received in revised form 27 April 2024; Accepted 18 May 2024
Available online 21 May 2024

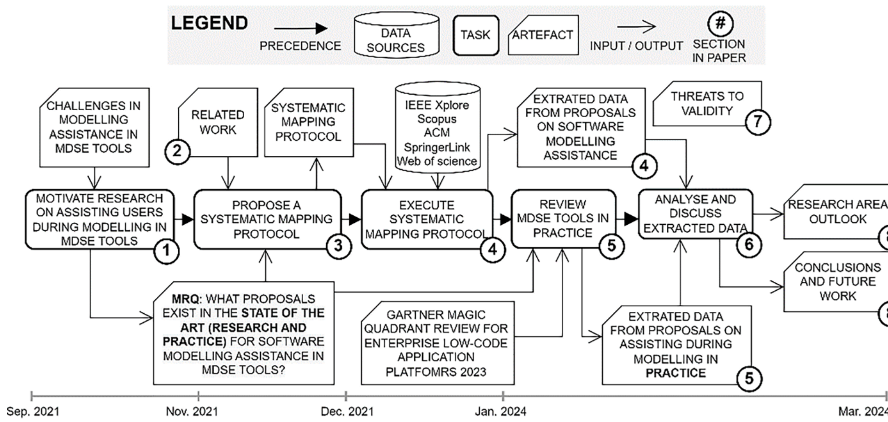

Fig. 1. Research overview.

are still maturing and improving modelling assistance is an identified not-yet-solved challenge [4]. Understanding the behaviour and skills of users, understanding the modelling context, and increasing the automation scope are examples of not-yet-solved challenges in modelling assistance. Given these open challenges in modelling assistance, it's crucial to explore how both literature and practice in MDSE have evolved to tackle them, aiming to enhance future solutions even further. However, how can we improve further solutions without knowing what is the state of development in this area? Other authors have asked the same question and motivated them to perform literature reviews on specific domains, technologies, and strategies [5–12]. Nevertheless, to the best of our knowledge, there is a lack of a review both in literature and practice without any restriction regarding technology, strategy, or domain on assisting users during modelling in MDSE tools. Therefore, in this paper, we propose the following main research question (MRQ):

***MRQ**: What proposals exist in the literature and practice to assist humans during modelling tasks in MDSE tools?*

From the perspective of literature, we conducted a systematic mapping study based on Petersen et al. [13] guidelines to gather data from the literature around our MRQ. We extracted data around i) how is software modelling assisted? ii) what goals and limitations do existing modelling assistance proposals report? and iii) which evaluation metrics and target users do existing modelling assistance proposals consider for modelling assistance? From the perspective of practice, we reviewed MDSE tools from the Gartner Magic Quadrant (GMQ) for Enterprise Low-Code application platforms [14]. We extracted data about: what is the state of the practice on modelling assistance?

To gather data from the literature, three reviewers selected 58 proposals from a set of 3.176 records using inclusion/exclusion criteria. Having 58 proposals selected with a substantial inclusion agreement [15]—i.e., having a $0.8 > K$-statistic > 0.61—we extracted the data to answer the MRQ. We clustered, analysed, and discussed the data.

Based on the results, we propose a set of clusters—i.e., we grouped proposals—regarding their strategies, goals, limitations, target users, and evaluation metrics to answer the proposed RQs. We provide definitions and examples for each cluster (see Section 4). We observed that some proposals mainly assist users during modelling tasks in MDSE tools using software-based strategies, including tools, techniques, methods, frameworks, and languages. In contrast, other proposals use no-software-based strategies such as guidelines. Regarding goals, we found seven clusters of goals such as addressing change propagation, enhancing consistency checking, ensuring model compatibility, improving model quality, improving user interaction, easing model evolution and supporting vulnerability detection. Thus, we observed that proposals aim to assist users in creating and refining existing models in MDSE tools. On the other hand, we identified five limitation clusters, including accuracy, effort, generality, learnability, and scope limitations. Similarly, we found three clusters for evaluation metrics, including effectiveness, efficiency, and user perception metrics. Regarding target users, we found three clusters: designers/modellers, domain experts, and software developers. However, we observed that several proposals have no specified limitations, evaluation metrics, and target users, hindering gap identification, further analysis, and arising inconsistencies.

To gather data from the practice, we explored the 17 MDSE tools that were mentioned in the GMQ [14] and extracted quotes about strategies, goals, limitations, target users, and evaluation metrics for assisting during modelling tasks. We observed that some MDSE tools have more than one strategy for assisting users during modelling tasks. Nevertheless, not all MDSE tools have documented their modelling assistants, hindering data extraction and comparison.

After analysing our results, we have provided an outlook on the modelling assistance research area. We provide a comparative analysis of our results, comparing clusters found in both literature and practice. Our research highlights a gap in software modelling assistance, especially in terms of explicitly defining limitations, evaluation metrics, and target users. Moreover, we anticipate disruptive changes in the modelling assistance strategies and goals with the rise of artificial intelligence-driven technologies in the following years. To face this disruption and close the identified gap, we consider proposing a unified framework essential for improving software modelling assistants' design. We reflect on this as future work and propose to align our results with existing frameworks for designing modelling assistants.

We expect researchers and MDSE practitioners to benefit from our results in designing new software modelling assistants, relying on existing goals and targeting existing gaps, avoiding mentioned limitations, considering listed evaluation metrics, and targeting specific users. As a final step, we'll discuss how our study aligns with our research agenda. We provide a complete overview of our research in Fig. 1.

This paper is structured as follows: In Section 2, we review the related works on systematic mappings about software modelling assistance; In Section 3, we show the systematic mapping study protocol; In Section 4, we explore the results from executing the systematic mapping study protocol; In Section 5, we conduct the review on MDSE tools available in practice based on the GMQ; In Section 6, we present a comparative analysis of our results; In Section 7; we discuss the threats to validity and limitations of our study; and, finally, in Section 8, we conclude our paper and point out future work.

2. Related work

In the literature, there are various systematic mapping and literature review studies that we grouped into two major groups: i) modelling assistance for Model-Driven Engineering tools (MDE)—i.e., proposals to assist in creating MDSE tools using an MDE tool like the Eclipse Modelling Framework [16]—and ii) specific strategies to assist users

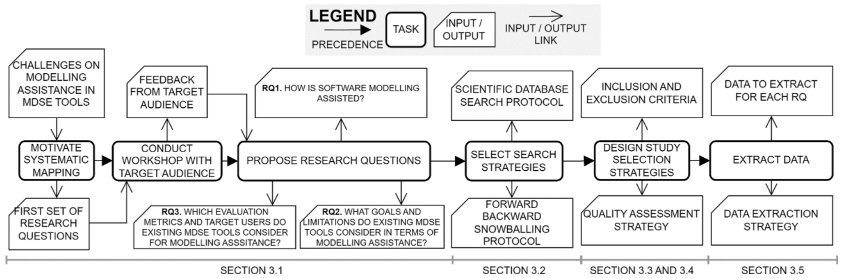

Fig. 2. Systematic mapping study design overview.

during modelling tasks in MDSE tools—e.g., reviewing proposals that use recommender systems as a strategy to assist users during modelling tasks in MDSE tools. We describe in detail such related works in the following paragraphs.

Group 1: Reviews on modelling assistance for developing MDE tools. The first group of related works aims to gather proposals around assisting modelling tasks from the model-driven engineering perspective. These studies review techniques, tools, and methods to assist model-driven engineers in creating MDSE tools, limiting the scope to MDSE engineers rather than MDSE tools' users. For instance, a systematic literature review on developing model transformations [5], a systematic mapping study on domain-specific language development tools [6,7], and a systematic literature review on testing model-to-text transformations [8].

Group 2: Reviews on modelling assistance for MDSE tools using specific strategies. The second group of related works aims to review proposals to assist users during modelling tasks in MDSE tools—i.e., proposals that assist MDSE tools' users in modelling software. These related works gather proposals with a common strategy to assist modelling tasks. Although this information is useful for understanding the existing proposals following a specific strategy, their analysis is restricted and lacks generality. For instance, a systematic mapping review on recommender systems in MDSE tools [9], a systematic literature review on elicitation techniques in MDSE [10], and a research map on collaborative MDSE [11].

Differentiating between both groups of related work is crucial for positioning our research. Our work diverges from the scope of the first group, as we do not focus on reviewing proposals aimed at assisting in the modelling of new MDSE tools. This decision is rooted in our consideration that creating tools for modelling new MDSE tools involves meta-modelling rather than modelling. In contrast, our research aligns with the scope of the second group of related work. While sharing a common scope, our research adopts a broader perspective, considering proposals in modelling assistance without restricting them to a specific strategy. Similarly, some authors have conducted a systematic mapping aiming to review modelling assistants in software engineering [12]. This related work shares a similar scope as ours since it focuses on proposals that aim to assist with software engineering tasks from a general perspective (i.e., they do not restrict their scope to a specific strategy). Nevertheless, their scope includes software engineering tasks rather than modelling with MDSE tools. Thus, to the best of our knowledge, we lack a review of modelling assistance literature and practice in MDSE tools without any restriction regarding a specific strategy. This gap motivates our study.

3. Systematic mapping study design

Systematic mapping studies are designed to overview a research area, searching the literature to know what topics have been covered [13]. We consider the following steps in our systematic mapping study: research questions, search strategy, inclusion/exclusion criteria, quality assessment, and data extraction. We deeply explore each step in the following sections. Moreover, we show graphically each step, including inputs and outputs, in Fig. 2.[1]

3.1. Research questions

In Section 1, we introduced the Main Research Question (MRQ) that motivates our study: *What proposals exist in the literature and practice to assist humans during modelling tasks in MDSE tools?*. Since this MRQ is highly generic, we needed to split it into a set of more specific research questions (RQs) related to a literature review. We proposed a set of RQs based on our experience with the topic: i) which kind of approaches are used for assisting modelling?; ii) which problem are they trying to solve?; iii) have they been empirically evaluated?; iv) how did they define the concept of the user? Based on Petersen et al. [13] guidelines, consulting the proposed RQs with the target audience is a good practice. Thus, we brought together nine software engineering experts from the Software Engineering research group at the ZHAW Zurich University of Applied Sciences. We consider them viable subjects to answer as the target audience of our study since they work on software-modelling-related topics. Based on the initial set of RQs and MRQ, we asked them: i) *Which RQ did you find more critical? and ii) What other RQ would you propose?* After discussing the results, we conceived the following three RQs:

RQ1. *How is software modelling assisted?* Answering this question, we aim to gather knowledge about the different strategies that have been developed to assist during modelling tasks in MDSE tools. For strategies, we expect to gather a set of tools, methods, techniques, and frameworks that have been proposed for modelling assistance. This can expand the understanding of modelling assistance proposals to promote further advancements in the field.

RQ2. *What goals and limitations do existing modelling assistance proposals report?* The goals and limitations of proposals give a deeper understanding of their potential impact and usefulness in a specific context. We expect to gather a set of goals and limitations to find gaps and existing trends.

RQ3. *Which evaluation metrics and target users do existing modelling assistance proposals consider?* Answering this question, we expect to collect data that shows which quality attributes are evaluated and to which target users. We acknowledge that MDSE tools have further functionalities than assisting software modelling. Therefore, we only consider target users that benefit from modelling assistance. On the other hand, evaluation metrics represent the quality attributes evaluated in existing proposals.

3.2. Search strategy

In our systematic mapping study, we combine two search strategies: database search and snowballing. First, we perform a database search in five scientific databases: IEEE Xplore, ACM Digital Library, Scopus,

[1] To facilitate replication of this systematic mapping study, all materials related to our research protocol can be found at https://zenodo.org/records/10262145.

Springer Link, and Web of Science. We require a search string to search in such scientific databases. Therefore, we use the PICO (Population, Intervention, Comparison and Outcomes) [17] to identify the keywords and formulate the search string as follows:

Set of terms comprising the Population. In our study, we consider population as MDSE tools. We exclude generic modelling tools—e.g., draw.io, diagrams.net, lucidchart.com—since our goal is analysing modelling assistants in tools that automatically produce software as an output. Therefore, we propose the following terms for describing our population: *Model-Driven Development (MDD), Model-Driven Engineering (MDE), Model-Driven Architecture (MDA), Model Driven Software Engineering (MDSE), Model-Based Software Engineering (MBSE), Low code, No-code.*

Set of terms related to the Intervention. In our study, we focus on proposals that aim to assist "users" during modelling tasks as intervention. We exclude any term related to a specific strategy or tool to assist such "users" during modelling tasks—e.g., recommender system, machine learning, rule-based assistant—since our goal is to gather as many proposals as possible, no matter the strategy. Moreover, we aim to include specific terms about the "user" as the intervention since we aim to gather proposals that have a human user. Therefore, we propose the following terms for representing our intervention: *assist, support, help, maintain, promote, ease, facilitate, simplify, user, developer, tester, software engineer, architect, usability, usage, approach, proposal, concept, idea, method, manner, technique, procedure, program, assistant.*

Then, we propose the following search string based on such a set of keywords:

Search string: *("Model-Driven Engineering"* ***OR*** *"MDE"* ***OR*** *"Model-Driven Architecture"* ***OR*** *"MDA"* ***OR*** *"Model Driven Software Engineering"* ***OR*** *"MDSE"* ***OR*** *"Model-Driven Development"* ***OR*** *"MDD"* ***OR*** *"Model-Based Software Engineering"* ***OR*** *"MBSE"* ***OR*** *"Low code" OR "No code")* ***AND*** *("support*"* ***OR*** *"assist*"* ***OR*** *"help*"* ***OR*** *"ease"* ***OR*** *"facilitate*"* ***OR*** *"simplify*")* ***NEAR TO*** *("user"* ***OR*** *"developer"* ***OR*** *"tester"* ***OR*** *"software engineer"* ***OR*** *"analyst"* ***OR*** *"architect" OR "usability" OR "usage")* ***AND*** *("approach"* ***OR*** *"proposal"* ***OR*** *"concept"* ***OR*** *"idea"* ***OR*** *"method"* ***OR*** *"manner"* ***OR*** *"technique"* ***OR*** *"procedure" OR "program"* ***OR*** *"assistant")*

We use the search string in five scientific databases applied to papers' titles and abstracts, adapting the search string to the specific database requirements—e.g., limiting the number of wildcards (*). We restrict the time range of our database search from 1985 to 2024 (the last year of execution). After executing the database search strategy, we use the procedure proposed by Wohlin [18] to implement the snowballing search strategy. First, we need to select an initial set of records for starting the snowballing. We use the inclusion/exclusion criteria (see Section 3.3) to select the first set of possible proposals from the database search. Then, we choose the top 12 records based on the quality assessment (see Section 3.4). Based on this first set of 12 records, we start iterating the references (backwards snowballing) and citing records (forward snowballing) in rounds using the inclusion and exclusion criteria. We stop iterating when no new records are found at the end of the round.

3.3. Inclusion/exclusion criteria

We propose the following set of inclusion/exclusion (I/E) criteria:

- **(I1)** Is the paper exclusively dedicated to a particular proposal, rather than being a compilation of proposals, aimed at assisting users during modelling tasks in MDSE tools? Compilation of proposals like literature reviews, systematic mappings, or systematic literature reviews does not fulfil I1.
- **(I2)** Is the proposal designed to assist users during modelling tasks in MDSE tools? We focus on proposals that assist users during modelling tasks in MDSE tools, including—but not limited to—modelling, model tracing, model debugging, model repair, and model validation, among others. As mentioned in Section 2, proposals specifically targeting the development of new MDSE tools do not fulfil I2. This is because creating a new MDSE tool involves meta-modelling rather than modelling, similar to the distinction between assisting in creating a modelling language and assisting in using a modelling language itself.
- **(E1)** The proposal's main contribution is not to assist users during modelling in MDSE tools. If assisting users during modelling tasks is not the main contribution, we exclude the proposal using E1—e.g., a proposal showing a new MDSE tool where assisting users during modelling tasks is superficially mentioned and is not the main goal of the proposal.
- **(E2)** The proposal is not related to software engineering.
- **(E3)** The proposal is not written in English.
- **(E4)** The proposal is not a peer-reviewed publication.
- **(E5)** The proposal's full text is not available.

Table 1
3-Point-Likert-scale questionnaire for quality assessment.

Subjective Questions	1 = Yes	0 = Partially	−1 = No
1. Is the proposal for assisting users during modelling tasks in MDSE tools clear?			
2. Are the proposal's limitations clear?			
3. Are the proposal's goals clear?			
4. Are the proposal's tools/sources downloadable?			
5. Is there a clear case study or example illustrating the proposal?			
6. Is the whole proposal empirically evaluated?			
7. Are the users clearly described?			
8. Are the results clearly explained?			
Objective Questions			
9. How important is the proposal's publication venue?			
	1 = Very important	0 = Important	−1 = Not that important
CORE Range:	A* - A - B	C	Not indexed
JCR Range:	Q1 - Q2	Q3 - Q4	Not indexed
10. How many citations do the proposal have?			
	1 = More than 4	0 = Between 2 and 4	−1 = Less than 2

Reviewers use I1-E5 to screen proposals' titles, abstracts, and full texts. Reviewers must agree that a proposal fulfils all inclusion criteria and does not fulfil any exclusion criteria to consider it as an included proposal.

3.4. Quality assessment strategy

Based on [19], we combine two types of quality assessments into a 3-Point-Likert-Scale questionnaire: subjective and objective quality assessments. For subjective quality assessment, reviewers answer questions based on their opinions about the proposal. For objective quality assessment, reviewers use the CORE ranking[2] for conferences, the Journal Citation Report[3] (JCR) for journals, and the proposal's number of citations. Reviewers directly extract the number of citations from each paper's database overview dashboard. In case of absence, reviewers use Google Scholar to extract the number of citations. We show the quality assessment questionnaire in Table 1. Quality assessment provides us with subjective and objective data to ensure the reliability and validity of the findings. Also, as we mentioned, we use the top 12 studies based on quality assessment results as the initial set of proposals for snowballing execution.

3.5. Data extraction strategy

We aim to extract data from our systematic mapping studies based on RQ1, RQ2, and RQ3. Reviewers must extract the text fragment(s) that contains possible answers for each RQ as follows:

[2] http://portal.core.edu.au/conf-ranks/
[3] https://www.scimagojr.com

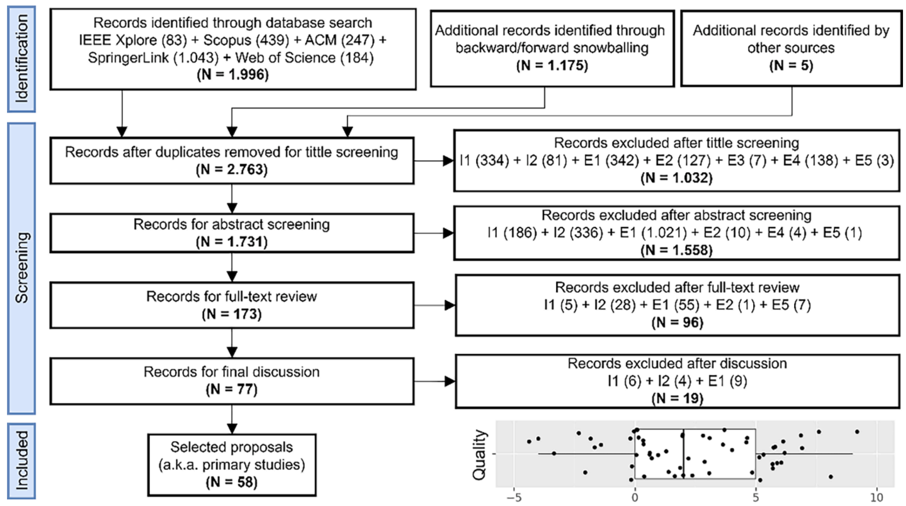

Fig. 3. PRISMA flow chart for proposal selection and quality assessment results.

- RQ1: How is software modelling assisted? Extract the keywords the proposals' authors use for describing the proposal's strategy to assist during modelling tasks—e.g., methodology, technique, framework, among others.
- RQ2: What goals and limitations do existing modelling assistance proposals report? Extract all goals and limitations the authors state the proposal has. If authors do not state any goal or limitation, leave the field blank.
- RQ3: Which evaluation metrics and target users do existing modelling assistance proposals consider? Extract if the authors empirically evaluate the proposals, which evaluation metrics they use for the evaluation, and which target users they expect to use their proposal. Leave the field blank if the authors do not state something about this information or if the authors use "user" to refer to their target users.

After extracting the literal data from the studies comprising RQ1, RQ2, and RQ3, we proceed to cluster and analyse them. We utilised the terminology used by the authors to create these clusters and carried out a cluster-level analysis of the data. It is important to note that this approach may introduce bias in data extraction and subjective interpretation. We extensively discuss these validity threats to our study in Section 7.

4. Systematic mapping study results

4.1. Systematic mapping study protocol execution results

Three reviewers were involved in the proposal selection—a.k.a. primary study selection. The first reviewer (R1) has more than four years of experience in model-driven-development-related scientific topics (first author of this paper). The other two reviewers (R2 and R3) are Computer Science MSc students with limited knowledge of model-driven development. However, we provide them with the necessary tools and training for conducting the proposal selection. Moreover, we introduce a fourth reviewer (R4) who has more than 10 years of experience in model-driven-development-related scientific topics (second author of this paper) to review the resulting clusters to mitigate the subjective interpretation validity threat (see Section 7.1).[4]

First, we executed the database search using the search string (see Section 3.2), resulting in an initial set of 1.996 records. We included five additional records suggested by external reviewers. Then, R1 and R2 reviewed the first 2.001 records using the inclusion/exclusion criteria (see Section 3.3), resulting in 51 possible proposals. R1 and R2 assessed the quality (see Section 3.4) of these 59 possible proposals and selected the top 12—i.e., possible proposals that fall above the 80th percentile based on the quality metric—for executing the snowballing search strategy (see Section 3.2). After four rounds of backward and forward snowballing, R1 and R2 reviewed 1.175 records more. In total, R1 and R2 reviewed 3.176 records resulting from the databases, snowballing, and additional records. As a result, R1 and R2 selected 77 possible proposals. Then, R3 and R4 exhaustively reviewed the set of 77 possible proposals to validate the selection that R1 and R2 did. R3, R4, and R1 discussed the inclusion/exclusion of the 77 possible proposals, resulting in a final set of 58 proposals [20–77]. To measure the reliability of agreement between reviewers, we measured the inclusion reliability using the K-Statistic [15,78] between the first and final proposal selection. As a result, we observe a K-statistic = 0.634, showing a substantial agreement based on Landis and Koch's interpretation of the K-statistic [15] (0.80 > K-statistic > 0.61). Then, R1 proceeded to cluster the extracted data for the accepted proposals. After the clustering, R4 reviewed the extracted data and proposed clusters reporting disagreements. We measure again the K-statistic between the first clustering and the resulting clustering after R4 review. We observe a K-statistic = 0.651, showing a substantial agreement based on Landis and Koch's interpretation of the K-statistic [15] (0.80 > K-statistic > 0.61) for the clustering result. Including several reviewers and the clustering raise some internal validity threats. We discuss such threats in Section 7.1. We summarise the proposal selection and the quality assessment results in Fig. 3. In the following sections, we provide an in-depth description of the resulting clusters for each RQ.

[4] The data extracted from proposals is mainly text, sometimes full paragraphs. Therefore, we have pre-processed these data to allow us to summarise the results and present them in the paper. Thus, the data presented in this paper are preprocessed for discussion and analysis. However, the clustering was performed using the complete data. Furthermore, for the sake of transparency, replicability, and confidence in our study, we have published the raw data in a public repository that can be consulted here: https://zenodo.org/records/10262145.

Table 2
RQ1: How is software modelling assisted? Identified clusters in our study.

RQ1: How is software modelling assisted?	
Clusters	Keywords
Tools	Domain modelling recommender systems [38,66,76], artificial intelligence-empowered software assistants [32,68], domain modelling bots [69,73], modelling assistants [70,74,77], modelling plugins [75], view manager with a meta layout [59], modelling environments [30], virtual reality environments [64], reactive systems [34], modelling environments [25,63,65], model testing tools [48,57], transformation-based tools [40,60], and collaborative management tools [29].
Guidelines	ISO-based standardisations [31], flexible workflows [41], refactoring processes [37], and multi-modelling architectures [26].
Techniques	Model development techniques [50,53], model validation techniques [22,42,44,45,51], and model repair techniques [35,55].
Methods	Consistency validation methods [24,28], service-oriented consistency-validation methods [21], model repair methods [20], task-driven methods for model reuse [36,39], and model-driven engineering methods for model alignment [56,62].
Frameworks	Agent-based change propagation frameworks [23,61], testing frameworks [67], collaborative modelling frameworks [47], co-evolution frameworks [43], formal frameworks [58], and modelling frameworks [27,52,54,71,72].
Languages	Mega-modelling languages [49], language extensions [46], and modelling templates [33].

Guidelines support to establish where, when, and how to assist during a modelling task (adapted from [79]). We found guidelines to: i) create models by standardising UML modelling based on the ISO42010 [31] and flexible workflows for user interface prototyping [41]; ii) refactor models using a transformation-based process [37]; and iii) trace models by using a multi-modelling architecture [26].

Techniques integrate predefined guidelines with selected tools to perform a task to assist modelling (adapted from [79]). Notice that ***Techniques*** do not function as a superset of ***Tools*** and ***Guidelines***. Instead, ***Techniques*** employ an integrated approach wherein guidelines and tools collaborate to assist modelling. We found techniques to: i) create models using user-artefact-centric [53] and concern-oriented [50] techniques; ii) validate models by using integrated-separated model comparison [51], data-flow testing [45], and model-to-model consistency comparison techniques [22,42,44]; and iii) repair models by using isolated-edition-based [55] and model-change-based consistency checking techniques [35].

Methods are a set of instructions that involve people, processes, and techniques to assist modelling (adapted from [79,80]). ***Methods*** integrate ***Techniques*** such as natural language processing and model-driven engineering, among other techniques. Nevertheless, ***Methods*** do not function as a superset of ***Techniques.*** Differently from ***Techniques, Methods*** cluster proposals that align people and processes with suitable

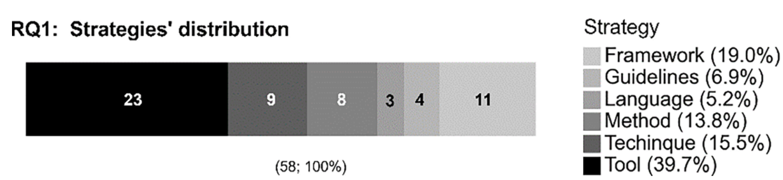

Fig. 4. RQ1: Strategies for modelling assistance distribution.

4.2. RQ1: how is software modelling assisted?

After analysing the extracted data for RQ1, we propose six clusters that emerged from the study and based on keywords from each proposal: *tools, techniques, methods, frameworks, guidelines, and languages*.[5] We also apply each cluster definition to classify the proposals. We show each cluster with the corresponding keywords in Table 2. Moreover, we show the distribution of strategies for assisting users during modelling tasks in MDSE tools in Fig. 4.

Tools are software implementations that perform a task to assist modelling (adapted from [79]). We found tools that perform tasks to assist modelling, such as: i) suggesting elements during modelling using domain modelling recommender systems [38,66,76], artificial intelligence-empowered software assistants [32,68], domain modelling bots [69,73], modelling assistants [70,74,77], and modelling plugins [75]; ii) managing model visualisation using view managers with a meta-layout [61], adaptable modelling environments [32], virtual reality environments [64], and reactive systems [34]; iii) resolving semantic discrepancies using modelling environments [25,63,65]; iv) checking consistency and model validity by testing models [48,57], and transformed-based graphs [40,60], v) creating models using automatic text-to-model transformations; and vi) support models synchronisation using a model collaboration tool [29].

techniques, producing a set of instructions to assist modelling. We found methods to: i) validate consistency using UML-based [28], natural language processing [24], and service-oriented methods [21]; ii) repair models using a method for generating repair plans [20]; iii) support model reuse and end-user involvement using task-driven methods [36,39]; iv) keep models aligned using model-driven engineering methods involving business processes [62] and heterogeneous models [56].

Frameworks are black boxes that include an overview of a set of elements and their relationship to assist modelling (adapted from [79]). Moreover, frameworks can be tool-supported. We found frameworks to: i) repair models using agent-oriented change propagation frameworks [23,61] and testing frameworks [67]; ii) edit models asynchronously using collaborative modelling frameworks [47]; and iii) evolve models by using co-evolution [43], formal [58], and modelling frameworks [27,52,54,71,72].

Languages are proposals that modify or contain a modelling language to assist modelling (adapted from [79]). We found languages such as: i) a mega-modelling language [49] to organise models; ii) a UML language extension [46] to verify pre and post conditions; and iii) a modelling template [33] to create models faster.

We observe *tools* are the most common strategy to assist during modelling (39.7 %), followed by *frameworks* (19.0 %), *techniques* (15.5 %) and *methods* (13.8 %). *Guidelines* (6.9 %) and *languages* (5.2 %) are the less common strategies to assist during modelling. Notice that we cluster each proposal in one cluster even if some overlap. For example, we classify a method that is tool-supported still as a method as long as it adjusts to the authors' keywords and cluster definitions for the *method*. Based on such distribution, we observed that 93.1 % of proposals assist during modelling using totally or partially software implementations—e.g., recommender systems (*tool*), model validation techniques (*technique)*, model repair methods (*method)*, agent-based change propagation frameworks (*framework)*, and modelling templates

[5] We recognise that in software engineering research, definitions of method, framework, technique, and tool are still not unified—i.e., an author can consider their proposal as a tool, and other authors can consider the same proposal as a technique. In our study, we rely on the keywords adopted by the proposals' authors and our definition to each cluster. However, we acknowledge this lack of common vocabulary as a threat to validity and possible future challenges in this area. We mention this in Section 7.

Table 3
RQ2: What goals (G) and limitations (L) do existing modelling assistance proposals report? Identified clusters in our study.

RQ2: What goals (G) and limitations (L) existing modelling assistance proposals have?	
Clusters	Keywords
G1. addressing change propagation	Understand how model changes affect the model correctness [51], allow asynchronous changes [47], verify refactoring pre/post conditions [46], apply refactoring rules [37], detect inconsistent model changes [25,35], support change-based process formalisations [29], achieve model change traceability [26,27,67], and fix propagated changes [23,61].
G2. enhancing consistency checking	Keep models aligned [62], check between models and implementation [57], inspect inconsistencies in models [45,55,65], ensure model intra/inter consistency [21,24,44,56], and formalise consistency between models [28].
G3. ensuring model compatibility	Ensure component-based software compatibility guarantees in open development and runtime environments [54].
G4. improving model quality	Improve model quality by model validation, verification, and exploration [42].
G5. improving user interaction	Enhance user interaction mechanisms [59,64], reduce users' cognitive load [53], alleviate editing modelling expressions [52], semantically lift low-level differences [63], organise models [49], make amenable formal verification [48], preserve user's creativity during modelling [41], guide users on modelling [34,73], reduce modelling complexity [31], allow end-users to cooperate from modelling stage [39], enhance modelling learning experience [70,72], and make model management user friendly [22].
G6. easing model evolution	Automatically generate models [30,40,50,58,69], support model co-evolution [43], support domain modelling [38,71,74], support model reuse [36,75], decrease modelling time [33], and suggest model elements during modelling [20,32,66,68,77].
G7. supporting vulnerability detection	Detect model security vulnerabilities [60].
L1. accuracy	Limitations on model automatic extraction accuracy [50,73], meaningful suggestions [68,69], and inconsistency resolution accuracy [35].
L2. effort	Proposals increase the effort on modelling maintenance [51], require modelling standardisation [31], and introduce additional manual modelling tasks [20].
L3. generality	Limitations on specific domains [44,60], specific refactoring types [46], specific training data [74], and modelling scenarios [38,45,67,76].
L4. *learnability*	Users have problems learning uncommon modelling languages [62], learning how to make decisions [61], and learning underlying technologies [33].
L5. *scope*	Lack of specific modelling constraints [23,55,57], modelling components [54], assistance features [64,66,71,75], and model types [28].
L6. *usability*	Lack of usability-related features [65].
L-NS	[21,22,24–27,29,30,32,34,36,37,39–43,47–49,52,53,56,58,59,63,70,72,77]

G: goal; **L:** limitation; **NS:** Not specified; **MDE:** Model driven engineering.

(language). In contrast, 6.9 % of proposals assist users to model software using *guidelines (6.9 %)*. *Guidelines* assist users during modelling, not necessarily using software, for instance, by standardising the steps users should follow to model. Therefore, we observe a trend in assisting users to model software using software-based strategies.

4.3. RQ2: what goals (G) and limitations (L) do existing modelling assistance proposals report?

After analysing the extracted data for RQ2, we propose seven clusters about proposals' goals and five clusters about proposals' limitations. Moreover, we classify a proposal as L-NS when authors do not explicitly specify their limitations. We show each cluster with the corresponding keywords in Table 3. In Fig. 5, we show a bubble chart comprising limitations and goals clusters, also having their distribution among proposals.

G1. Addressing change propagation. We found proposals aiming to address change propagation during modelling tasks in MDSE tools. We frame change propagation as an umbrella comprising model synchronisation, change-based inconsistency detection, and change traceability during modelling tasks. For example, a proposal [51] aims to assist users during modelling tasks addressing change propagation by detecting how correct the models are when a change is introduced. Similarly, we cluster all proposals that aim to assist in easing, explaining, detecting, tracing, fixing, and verifying changes during modelling tasks in G1.

G2. Enhancing consistency checking. We found proposals aiming to enhance consistency checking during modelling tasks in MDSE tools. We observe that consistency checking during modelling tasks in MDSE tools involves not only inter-consistency—i.e., consistency between models inside MDSE tools—but also external consistency with other models, code snippets, or requirements. Differently from G1, proposals in G2 aim to enhance intra/external consistency checking between models after changes have been already propagated—i.e., independently on how change propagation is addressed. For example, a proposal [55] aims to assist during modelling tasks in MDSE tools by detecting inconsistencies between models introduced by isolated editing of single views. Similarly, we cluster all proposals that aim to assist in checking, detecting, inspecting, showing, and ensuring consistency checking during modelling tasks in G2.

G3. Ensuring model compatibility. We found a proposal [54] aiming to ensure model compatibility during modelling tasks in MDSE tools. We observe that model compatibility is relevant when the resulting model is used as a component-based software with other open development and runtime environments interacting with it. We cluster proposals that aim to assist in creating, generating, and producing compatible models in G3.

G4. Improving model quality. We found a proposal [42] aiming to improve model quality during modelling tasks in MDSE tools. We consider "model quality" as a broad term. In this proposal, we found model quality as a mix of verification, validation, and exploration. Notice that G4 do not function as a superset of other goal clusters with proposals that aim to assist on improving a model quality aspect—e.g., proposals that improve model consistency in G2 by enhancing consistency checking. Instead, proposals in G4 aim to assist on improving more than one model quality aspect. Thus, we cluster proposals that aim to assist with more than one quality aspect during modelling tasks such as validation, verification, and exploration in G4.

G5. Improving user interaction. We found proposals aiming to improve user interaction during modelling tasks in MDSE tools. We observe that user interaction with MDSE tools involves how users effectively use MDSE tools to achieve their goals. For example, a proposal [63] aims to assist users during modelling tasks, improving user interaction by providing user-comprehensible specifications of changes between two models. Similarly, we cluster all proposals that aim to assist in alleviating, easing, supporting, and enhancing user interaction during modelling tasks in MDSE tools in G5.

G6. Easing model evolution. We found proposals aiming to ease model evolution in MDSE tools. We observe that model evolution involves implementing model instances, creating models, co-evolving models based on historical data, or editing existing models. For instance, a proposal [20] aims to assist users during modelling tasks by easing model evolution and generating repair plans when an error/bug is triggered. Similarly, we cluster all proposals that aim to assist in supporting, implementing, and generating model evolutions in G6.

G7. Support vulnerability detection. We found a proposal [60] aiming to support vulnerability detection during modelling tasks in MDSE tools. We observe that vulnerability detection is relevant for ensuring that the resulting software complies with non-functional requirements related to security and privacy. This proposal's goal is to detect security vulnerabilities and suggest corrective actions to address these weak points during modelling tasks. Thus, we cluster proposals that aim to assist in

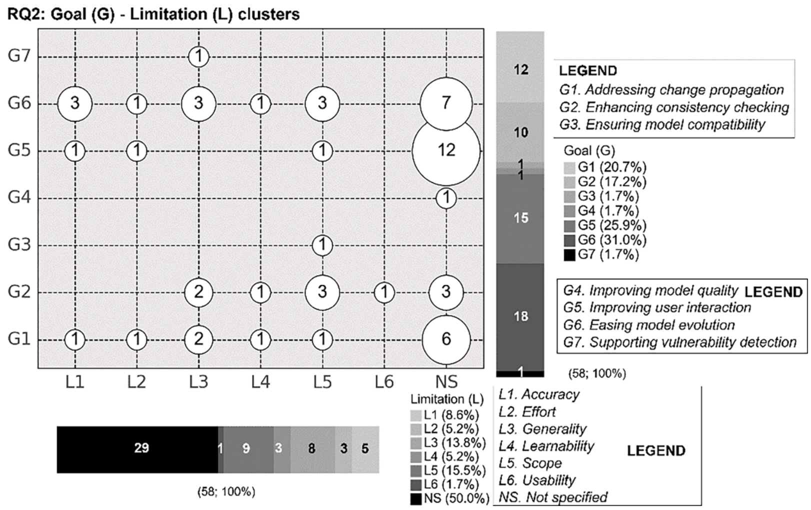

Fig. 5. RQ2: Goal (G) – Limitation (L) bubble chart and distributions.

identifying, correcting, and detecting vulnerabilities during modelling tasks in MDSE tools in G7.

L1. Accuracy. We found proposals reporting accuracy limitations. We cluster proposals in L1 when the authors state the obtained results are not the expected result. For instance, a proposal [50] reports accuracy limitations on the results of automatic model extraction. We cluster all proposals that report a lack of accuracy in L1.

L2. Effort. We found proposals reporting effort limitations. We cluster proposals in L2 when the authors state their proposal requires increasing effort before, during, and after modelling tasks. For example, a proposal [31] reports effort limitations since their proposal requires previously standardising all the MDSE tool architecture metamodels. We cluster all proposals that report similar effort limitations in L2.

L3. *Generality*. We found proposals reporting generality limitations. We cluster proposals in L3 when the authors acknowledge that—even if their work contributes a novelty and advances state of the art—their resulting proposal is limited to specific modelling scenarios—i.e., even if the obtained results meet what authors expected, the generalizability of these results is limited. For instance, some proposals [44,60] report generality limitations due to their proposal being limited to a specific domain. We cluster proposals that report similar generality limitations in L3.

L4. Learnability. We found proposals reporting learnability limitations. We cluster proposals in L4 when the authors state their proposal requires users to learn extra skills, independently if mastering these new skills ultimately affects the overall effort during modelling. For example, a proposal [62] reports learnability limitations due to users' need to learn a non-well-known modelling language in a business environment. We cluster all proposals that report similar learnability limitations in L4.

L5. Scope. We found proposals reporting scope limitations. We cluster proposals in L5 when the authors acknowledge that their proposal lacks specific features to achieve their goal. For instance, some proposals [23,55,57] report scope limitations due to their lack of support for specific consistency constraints during modelling tasks. We cluster all proposals that report similar scope limitations in L5.

L6. Usability. We found a proposal reporting usability limitations. We cluster proposals in L6 when the authors recognise that their proposal's usability is yet to be assessed or can be improved. For instance, authors from [65] report that their proposal's user interface needs to be improved by having extra support for users—i.e., including a guided step-by-step process.

Based on what we observe, existing proposals mostly assist users in refining existing models in MDSE tools. In total, 31.0 % (18) proposals aim to assist users specifically in creating models by *easing model evolution (G6)*. On the other hand, many proposals aim (25, 43.1 %) to refine already existing models by *addressing change propagation (G1), enhancing consistency checking (G2), ensuring model compatibility (G3), improving model quality (G4), and supporting vulnerability detection (G7).* Finally, there are proposals (15, 25.9 %) that aim to assist users in both creating and refining models by *improving user interaction (G5)* in MDSE tools.

Moreover, we observe that many proposals do not specify their limitations, which hinders the identification of research gaps. Half of the proposals (29, 50.0 %) explicitly specify *limitations.* This also raises inconsistencies. For instance, many proposals aiming to *improve user interaction (G5)* do not specify any limitations. Even though there are still many open challenges related to *improving user interaction* [81].

Table 4
RQ3: Which evaluation metrics (M) and target users (U) existing modelling assistance proposals consider? Identified clusters in our study.

RQ3: Which evaluation metrics (M) and target users (U) existing modelling assistance proposals consider?	
Clusters	Keywords
M1. effectiveness	Number of detected real faults [57], the degree of stakeholder participation [62], F-measure [66], feasibility of execution [56], inconsistency coverage [55], success scores [53,66,74], accuracy [20,40,51,68], compression factor [63], collection of execution traces [67], effectiveness [61,69], recall [66,68,71,74], accepted suggestions [71], and precision [24,66,71,74].
M2. *efficiency*	Modelling time [53,59], model completion time [58], testing generation time [57,66], model repair generation time [55], performance [52,60,61,63,69,71], computational effort [51], recommendation time [66], preprocessing time [66], resource import in seconds [75], reduction in number of executions [67], and execution time [20,67,74].
M3. *user perception*	Perception of industrial adoption [56] and perceived usefulness [29,69].
NE	[21–23,25–28,30–39,41–50,54,64,65,70,72,73,76,77]
U1. *designers/ modellers*	Software designers [61,71], model developers [58], professional engineers with design experience [56], designers [31], modellers [38,44,66,68,69,73,74], student modeller [70], novice modeller [72], UML developers [20], and MDE developers [48].
U2. *domain experts*	Business analysts [62], end-users [25,39], domain users [59], domain experts [41,65], domain engineers [65], and business users [29].
U3. *software developers*	Developers [24,26,30,33,35,40,42,43,46,47,49,50,55,76], software developers [54], software engineering students [53] and software maintainers [23].
U-NS	[21,22,27,28,32,34,36,37,45,51,52,57,60,63,64,77]

M: metric; **U:** Target user; **NS:** Not specified; **NE:** Not evaluated; **MDE:** Model driven engineering.

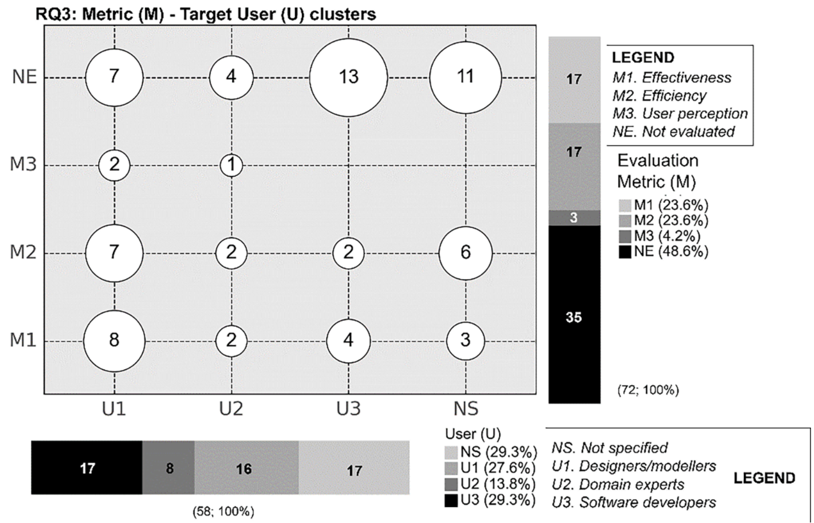

Fig. 6. RQ3: Metric (M) – Target User (U) bubble chart and distributions.

Regarding the relationship between goal clusters and limitation clusters, we observe that *G2 (enhancing consistency checking)* shows a relevant relationship, where half (5; 50.0 %) of the proposals report *generality (L3)* and *scope (L5)* limitations. On the other hand, we observe that other proposals are equally distributed over the limitations, such as those classified in *G1 (addressing change propagation)* and *G6 (easing model evolution)*. *G7 (supporting vulnerability detection)* and *G3 (improving model quality)* contain just one proposal each, limiting the analysis. Finally, as we mentioned before, *G5 (improving user interaction)* and *G4 (improving model quality)* have many proposals without explicitly specifying limitations.

4.4. RQ3: which evaluation metrics (M) and target users (U) do existing modelling assistance proposals consider?

After analysing the extracted data for RQ3, we propose three clusters about proposals' metrics and three clusters about proposals' target users. In the case of metrics, we classify the metrics we found using the Technology Acceptance Model (TAM) [82]. Notice that a proposal could have more than one evaluation metric specified. For users, we cluster them based on the extracted text. Moreover, we cluster proposals that do not specify an evaluation metric when they were not evaluated (NE). On the other hand, we cluster proposals as user-not-specified (U-NS) when authors use the generic keyword "user" or refer to "he/she" as the intended user to assist with their proposal. We show each cluster with the corresponding keywords in Table 4. In Fig. 6, we show a bubble chart comprising evaluation metrics and target users, also having their distribution among proposals.

M1. Effectiveness. Based on TAM [82], effectiveness metrics intend to measure the degree to which a proposal achieves its objectives. Thus, we cluster all proposals that explicitly specify metrics that fit with that definition in M1. For example, a proposal [57] that aims to assist in checking where there is an inconsistency between models and code proposes using the "number of detected real faults" as an evaluation metric. We cluster this metric into M1 since the metric reflects how effective the proposal is in detecting inconsistencies and faults between model and code—i.e., the proposed metric shows the degree to which the proposal meets its objectives.

M2. Efficiency. Based on TAM [82], efficiency metrics intend to measure the effort to apply a proposal. Thus, we cluster all proposals that explicitly specify metrics that fit with that definition in M2. For instance, some proposals [53,59] use as an evaluation metric the "modelling time." We cluster this metric into the M2 since "modelling time" is a metric that represents how much effort a user needs to complete a modelling task.

M3. User perception. Based on TAM [82], user perceptions are divided into perceived ease of use, usefulness, and intention to use. Then, user perception metrics can measure the degree to which a person believes that using a particular proposal would be free of effort; the degree to which a person believes that a particular proposal will be effective in achieving its intended objectives; or to what extent a person intends to use a particular proposal. Thus, we cluster all proposals that explicitly specify metrics that fit with one or more of such definitions in M3. For example, a proposal [56] uses "perception of industrial adoption" as an evaluation metric. We cluster this metric into M3 since "perception of industrial adoption" measures to what extent a person in the industry would adopt the proposal.

U1. Designers/modellers. We found proposals that consider *designers or modellers* as their target users. We cluster proposals into U1 when their target users' keywords refer to a person who specifies models/designs in MDSE tools. For example, a proposal [58] states that their target users are "model developers." We cluster this proposal into U1 since model developers' main goal is developing models in an MDSE tool.

U2. Domain experts. We found proposals that consider *domain experts* as their target users. We cluster proposals into U2 when their target users' keywords refer to a person who has domain expertise in the modelled business but not necessarily in modelling. For instance, a proposal [29] states that their target users are "business users." We cluster this proposal into U2 since business users have expertise in their business domain—e.g., healthcare, finances, insurance—but not necessarily in modelling.

U3. Software developers. We found proposals that consider *software developers* as their target users. We cluster proposals into U3 when their target users' keywords explicitly contain the keyword *developer* and/or involve a role relating to software engineering that does not fit in U1 or U2. For example, a proposal [54] states that its target users are "software developers." We cluster this proposal into U3 since their target user explicitly contains the word "developer" and does not specify a role that fits in U1 or U2.

Based on what we observe, many proposals have been evaluated using objective metrics, leaving aside subjective user perceptions. The most common evaluation metrics (34; 47.2 %) around proposals are effectiveness (M1; 17; 23.6 %) and efficiency (M2; 17, 23.6 %). This shows a trend toward measuring objective evaluation metrics—i.e., metrics that do not require direct input from the user. On the other hand, user perception metrics (M3) are a minority (3; 4.2 %). However, many proposals define no metric/were not evaluated (35; 48.6 %). In terms of target users, we observe *software developers* (U3) are the most common

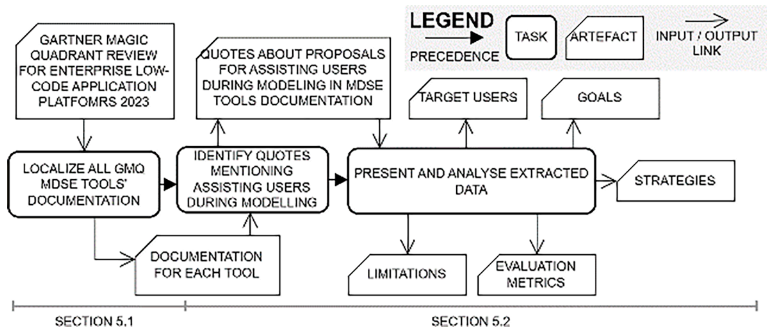

Fig. 7. Overview of reviewing MDSE tools.

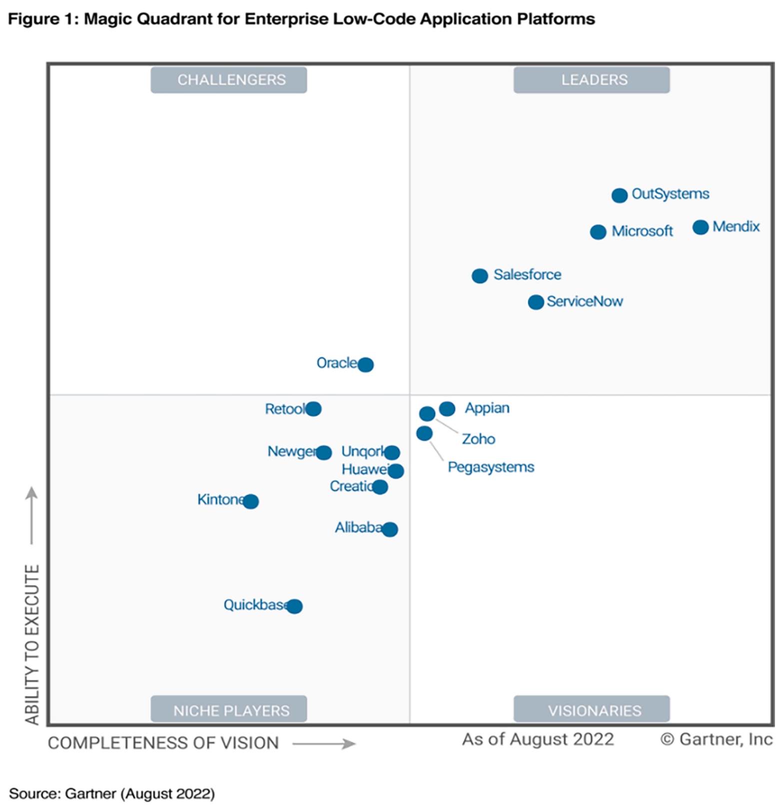

Fig. 8. Gartner Magic Quadrant for enterprise low-code application platforms. Source: [14].

target users among proposals (17; 29.3 %), followed by *designers/modellers* (U1; 16; 27.6 %). In contrast, we observe *domain experts* (U2; 8; 13.8 %) are the less common target users among proposals. Finally, we notice that there are many proposals (17; 29.3 %) that do not specify any target user.

5. State of the practice on modelling assistance in MDSE tools[4]

Up to this point, we have explored studies in the literature on modelling assistance based on a systematic mapping study. This view is relevant, as it is research that guides innovation in the industry. On the other hand, MDSE is a widely practical field. This implies that the literature alone does not contain all current advancements in modelling assistance. Therefore, we consider it relevant to review the state of the practice on how MDSE tools have been proposed to assist their users during modelling tasks. Thus, we introduce the following research question (RQ):

- **RQ4.** *What is the state of the practice on modelling assistance?* To answer this question, we aim to gather data from MDSE tools used in practice. Thus, we expect to extract descriptions of proposals for assisting during modelling in such MDSE tools. This includes strategies, goals, limitations, target users, and evaluation metrics.

First, we plan to explore which MDSE tools are available in practice to answer RQ4. To do so, we rely on Gartner Magic Quadrant (GMQ) for enterprise low-code application platforms [14]. GMQ is a culmination of research in a specific market, giving a wide-angle view of the relative positions of the market's competitors [83]. We explored all MDSE tools mentioned in the GMQ, looking for their documentation. Then, we review their documentation, extracting quotes about how they propose to assist users during modelling tasks. We show graphically these steps, including inputs and outputs, in Fig. 7.

Table 5
RQ4: What is the state of the practice on modelling assistance? Strategies (S), goals (G), limitations (L), evaluation metrics (M), and target users (U) based on quotes from GMQ MDSE tools' documentation.

MDSE Tool	Modelling assistance keyword	Quotes from documentation (i.e., websites, user guides, and technical documentation)
Mendix (LE)	MXAssist Logic Bot	"…Help development teams model and deliver Mendix applications faster…" (G6) "…Recommends the top five next best actions out of more than 40 different options…" (S:Tool) "…continuous improvement and training of the model has elevated the accuracy level to 95%…" (M1) "…when the developer inserts a new element…" (U3)
	MxAssist Performance Bot	"…assists you in improving the performance of your app by inspecting your app project model against Mendix development best practices…" (G4) "…the bot inspects the model, identifies an issue, and pinpoints the document/element causing the issue…" (S:Tool) "…the bot explains the identified issue's potential impact and shows the way to fix it…" (S:Tool) "…MxAssist Performance Bot enhances development efficiency by substantially reducing peer reviews…" (M2) "…increases developer productivity via the automatic detection and pinpointing of issues…" (M2) "…educates junior developers on best practices…" (U3)
	Validation Assist	"…helps you build validation microflows in a more automated way using pre-built expressions…" (G6) "…List of checks for all members which data type can be empty…" (S:Tool) "…Auto-generation of the validation microflow…" (S:Tool)
OutSystems (LE)	AI Code Mentor	"…assistant that guides you through the OutSystems platform, dramatically accelerating and improving application development.." (G6) "…this AI-powered capability is especially handy when you know which data you want to get from the database but are unsure how to add it to an aggregate…" (S:Tool) "…The feature may not work when you have a big number of entities and attributes…" (L3) "…You can only ask for data using English…" (L3) "…Typos in the name of entities, attributes, and variables may cause errors in the generated aggregate…" (L1) '…You can't ask for data filtered with advanced Functions…" (L5) '…You can't ask for data ordered using dynamic sorts…" (L5)
	Application templates	"…Thanks to the application templates, the apps have many predefined elements that save you development time…" (G6) "…Forge application templates…" (S:Tool) "…save you development time…" (M2)
	Architecture Mentor	"…AI Architecture Mentor makes sure your code meets critical architectural standards…" (G2)
	AI Security Mentor	"…AI-based security lead on your team to constantly review code to check for vulnerabilities…" (G7)
	AI Performance Mentor	"…Wishing you had a performance expert on your team to constantly review code, ensuring that apps will consistently perform and scale at peak levels? Now you do…" (G4)
	AI Maintainability Mentor	"…reduce technical debt and ensure best practices are followed…" (G2) "…scans your entire app portfolio…" (S:Tool)
Microsoft Power Apps (LE)	Copilot	"…Create Power Apps with the help of AI…" (G6) "…You can tell the AI assistant what kind of information you want to collect, track, or show, and the assistant will generate a Dataverse table and use it to build your canvas app…" (S:Tool) "…Preview features aren't meant for production use and may have restricted functionality…" (L3)
Salesforce (LE)	Performance Assistant	"…Use the step-by-step instructions, articles, and tools to help you architect your system, conduct performance testing, and interpret your results…" (G4) "…central hub of information and resources about scalability and performance testing with Salesforce…" (S:Tool)
	Einstein GPT for Salesforce Developers	"…automating the creation of repetitive code elements helps ensure consistency and standardisation in the codebase…" (G4) "…this not only speeds up the development process but also reduces the risk of human error…" (M1 and M2) "…generative code generation can speed up the prototyping process by quickly creating boilerplate code…" (S:Tool) "…it can save developers a lot of time…" (U3)
Appian (V)	Appian Developer Co-Pilot	"…helps developers while building process models by suggesting the nodes that we predict they are most likely to use next…" (G6) "…To help you, Appia" has added" "Suggested" nodes…" (S:Tool) "…That AI then helps developers…" (U3)
OrIe APPEX (C)	Intelligent Wizard	"…guide you through the rapid creation of applications and components…" (G6) "…Intelligent wizards to guide…" (S:Tool)
Retool (NP)	AI Features	"…help you write and edit SQL queries using natural language…" (G6) "…Retool's AI operates in three modes: generate, edit, and explain…" (S:Tool) "…Retool cannot guarantee that the output is always accurate…" (L1)
Not assistant found	ServiceNow (LE), Zoho Creator (V), PegaSystems (V), Newgen (NP),	

(continued on next page)

Table 5 (*continued*)

MDSE Tool	Modelling assistance keyword	Quotes from documentation (i.e., websites, user guides, and technical documentation)
	Unqork (NP), Huawei (NP), Creatio (NP), YiDA by Alibaba (NP), Kintone (NP), Quickbase (NP)	

LE: Leaders; **V**: Visionaries; **C**: Challengers; **NP**: Niche Players; **S**: Strategy; **G**: Goal; **L**: Limitation; **M**: Evaluation metric; **U**: Target user.

5.1. MDSE tools' review: gartner magic quadrant for enterprise low-code application platforms

Based on the GMQ for enterprise low-code application platforms in 2023 (see Fig. 8), there were 17 tools classified into challengers (1), leaders (5), niche players (8), and visionaries (3). In the following paragraphs, we introduce each cluster and its meaning based on GMQ, followed by a list of the MDSE tools classified in such clusters. In our study, we use GMQ as a research-supported and industry-relevant market study to gather grey literature on the documentation of MDSE tools. However, we acknowledge that there is other grey literature that could have been included but were not, such as reports and white papers. We mention this as a validity threat in Section 7.

Leaders (LE). MDSE tools classified as *leaders* execute comparatively well today, and they are well-positioned for tomorrow in the market [83]. We find five MDSE tools in such classification: OutSystems [84], Mendix [85], Microsoft Power Apps [86], Salesforce [87], and ServiceNow [88].

Challengers (C). MDSE tools classified as *challengers* execute comparatively well today or may dominate a large segment but do not have a roadmap aligned with Gartner's view of how a market will evolve [83]. We find just Oracle APEX [89] classified as a *challenger.*

Visionaries (V). MDSE tools classified as *visionaries* understand where the market is going or have a vision for changing market rules but do not yet execute comparatively well or do so inconsistently [83]. We find three MDSE tools classified as *visionaries:* Appian [90], Zoho Creator [91], and PegaSystems [92].

Niche players (NP). MDSE tools classified as *niche players* focus comparatively successfully on a small segment or are unfocused and do not out-innovate or outperform others [83]. We find eight MDSE tools classified as *niche players:* Retool [93], NewGenONE [94], Unqork [95], Huawei Astro Zero [96], Creatio ONE [97], YiDA by Alibaba [98], Kintone [99], and Quickbase [100].

5.2. RQ4: what is the state of the practice on modelling assistance?

We explore documentation, websites, and user guides from each MDSE tool mentioned in each GMQ 2023 cluster—i.e., leaders (LE), visionaries (V), challengers (C), and niche players (NP)—to answer RQ4. We search for insights into this data about how they aim to assist users during modelling tasks in such tools. Then, we extract the text and classify them following the same relevant data we extracted from RQ1,

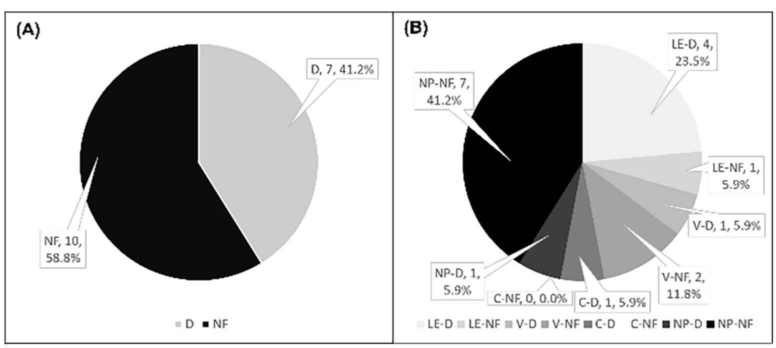

Fig. 9. (**A**) Distribution of GMQ MDSE tools with documentation on assisting during modelling (D) and not documentation found (NF). (**B**) Distribution of GMQ MDSE tools with documentation on assisting during modelling (D) and no documentation found (NF) per GMQ classification: Leaders (LE); Visionaries (V); Challengers (C); and Niche Players (NP).

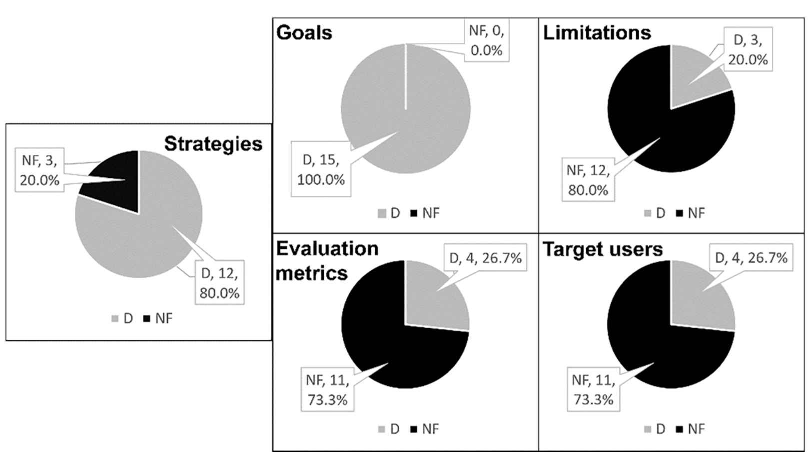

Fig. 10. Distributions on MDSE proposals for assisting users during modelling that mention (D) strategies, goals, limitations, evaluation metrics, and target users and which ones do not (NF).

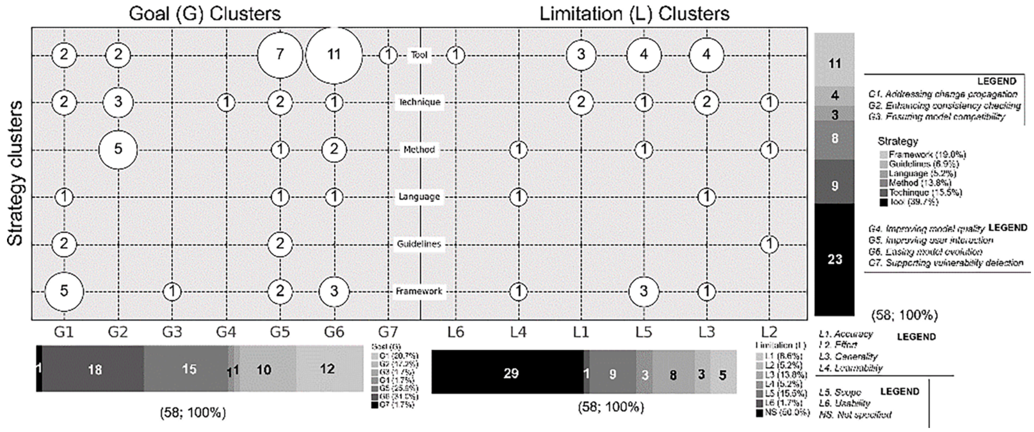

Fig. 11. Bubble chart from RQ1 and RQ2 clusters: Strategies, goals, and limitations.

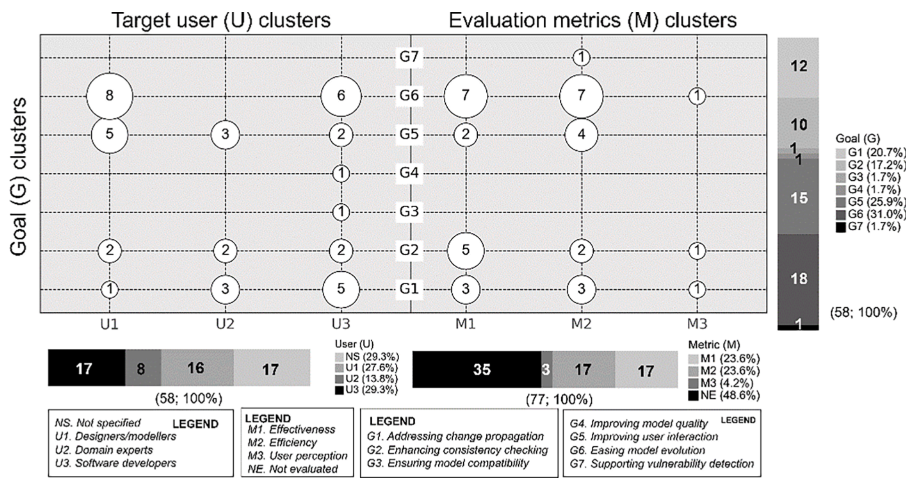

Fig. 12. Bubble chart from literature clusters from RQ2 and RQ3: Goals, evaluation metrics, and target users.

RQ2, and RQ3—i.e., we classify the text if we consider that the authors are referring to a strategy (S), goal (G), limitation (L), evaluation metric (M), or target user (U) cluster. In Table 5, we show the quotes from MDSE tools' documentation for each GMQ MDSE tool and how we classify them.

After reviewing seventeen MDSE tools we found in the GMQ, we observe that many of them do not mention how they intend to assist users during modelling in their documentation (10; 58.8 %; see Fig. 9A). In addition, of the seven tools (58.7 %) we observed that mention how they assist their users during modelling tasks in the documentation, many of them are part of the market leaders (LE; 4; 23.5 %; see Fig. 9B) according to the GMQ. On the other hand, at least one MDSE tool in each GMQ mentioned how they intend to assist users during modelling tasks in their documentation (see Fig. 9B).

Furthermore, we found fifteen proposals for assisting users during modelling inside the seven MDSE tools' documentation. In Fig. 10, we show the distribution of proposals mentioning strategies, goals, limitations, evaluation metrics, and target users.

We observe that many proposals (12; 80.0 %) mention the strategy they follow for assisting during modelling, including keywords such as: AI-power capabilities, templates, scanners, recommenders, model inspectors, AI assistants, and intelligent wizards. Notice that the strategies in these proposals are identified as tools. In terms of goals, we observe all proposals (15; 100.0 %) mention at least one goal they want to achieve by assisting users during modelling tasks, including keywords such as: improving application development, accelerating application development; ensuring code meets critical architectural standards; check for vulnerabilities; ensuring performance; validate models; create models; and speed up the prototyping process. In contrast, many proposals do not specify limitations (12; 80.0 %). The proposals that state limitations are a minority (3; 20.0 %), mentioning limitations such as: preview features are not meant for production, or the tool cannot guarantee that the output is always accurate. Regarding evaluation metrics, many proposals have not specified any metric (11; 73.3 %). Specified evaluation metrics among proposals include development time, accuracy, efficiency, productivity, and risk of human error. Similarly, many proposals have not defined a target user (11; 73.3 %). Specified target users include developers and junior developers. It is important to mention that due to the way the authors write their documentation, it is difficult to identify a specific target user. We found that authors write MDSE tools' documentation using the second person instead of the third person—i.e., authors write using the subject "you." This hides the actor performing the action—i.e., hides the target user. In Section 6, we provide a comparative analysis considering the clusters of RQ1, RQ2, and RQ3 we found in RQ4.

6. Comparative analysis: clusters in literature and practice

We have clustered proposals from literature and practice in Sections 4 and 5. In this section, we provide a comparative analysis between

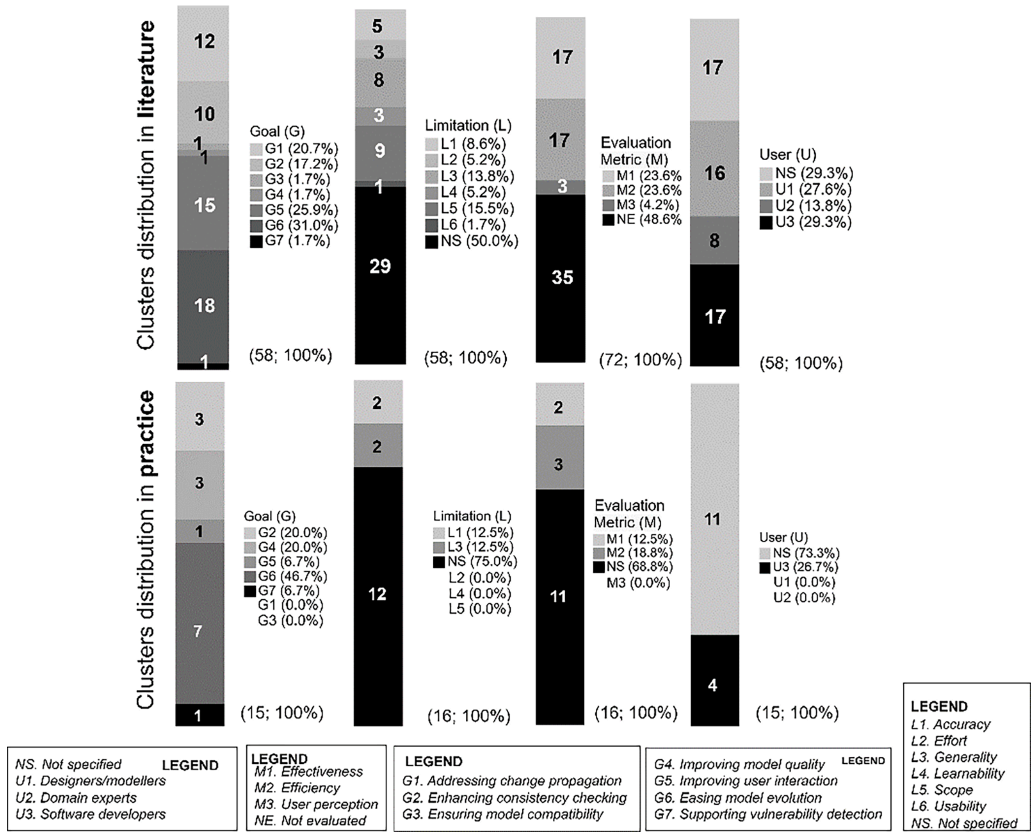

Fig. 13. Clusters distribution: literature vs practice.

clusters from RQ1 to RQ4.[6,7]

Modelling assistance strategies in the literature compared with goals and limitations. We observe in Fig. 11 that *Tools* have a predominance for addressing G5 (Improving user interaction) and G6 (Easing model evolution). Moreover, *Tools* show a similar distribution of limitations on accuracy (L1), scope (L5), and generality (L3). Regarding *Techniques,* we observe they are similarly distributed for most goals and limitations. In contrast, authors often propose *Methods* to enhance consistency checking (G2). We observe that *Languages* have been used to address change propagation (G1), improve user interaction (G5), and ease model evolution (G6). Moreover, we observe that *Guidelines* are used to address change propagation (G1) and improve user interaction (G5). We observe no clear distribution regarding limitations of *Methods, Languages, and Guidelines* due to the scarcity of data. Finally, we observe that *Frameworks* are present in most of the goals (G1, G3, G5, and G6). Nevertheless, authors use *Frameworks* especially to address change propagation (G1). On the other hand, we observe that *Frameworks* are mostly associated with scope (L5) limitations.

Modelling assistance goals in the literature compared with target users and evaluation metrics. We observe in Fig. 12 that many proposals aiming to ease model evolution (G6) consider designers/modellers (U1) and software developers (U3) as target users. Moreover, authors prefer to evaluate their proposals clustered in G6 using effectiveness (M1) and efficiency (M2) metrics with a similar distribution. Proposals aiming to improve user interaction (G5) have a similar distribution among target users, with a preference for targeting designers/modellers (U1). Moreover, we observe that proposals in G5 are evaluated using efficiency-related metrics (M2). Regarding proposals aiming to enhance consistency checking (G2), we observe that they target designers/modellers (U1), domain experts (U2), and software developers (U3) equally. We also observe that proposals in G2 are evaluated mostly using effectiveness (M1) related metrics. In contrast, many proposals aiming to address change propagation (G1) target software developers (U3) and domain experts (U2). Similarly, as proposals in G6, proposals in G1 use effectiveness (M1) and efficiency (M2) metrics equally for conducting evaluations. Due to the scarcity of data, we do not clearly observe distributions regarding G7, G4, G3, and M3.

Modelling assistance clusters: literature vs practice. In Fig. 13, we observe the distributions among clusters in both literature and practice. Regarding goals, we observe literature and practice coincide with many proposals aiming to ease model evolution (G6). In contrast, many proposals in the literature focus on improving user interaction (G5) and addressing change propagation (G1), while G1 and G5 are scarcely mentioned in practice. Regarding limitations, we observe both in literature and practice that many proposals do not define limitations. Moreover, we observe that in practice, accuracy (L1), generality (L3), and limitations are mentioned in a similar proportion to the literature. However, effort (L2), learnability (L4), and scope (L5) limitations are not mentioned in practice. Regarding evaluation metrics, we observe that both in literature and practice, evaluation metrics are often not mentioned. Nevertheless, in practice, we observe a predominance of not-specified evaluation metrics compared to the literature, having no mention of user perception metrics (M3). Finally, regarding target users, we observe that all proposals in practice target software developers (U3) as users. Thus, designers/modellers (U1) and domain experts (U2) are scarcely mentioned in practice compared to the literature.

[6] We remove Not-Specified/Not-Evaluated clusters from the bubble charts to remove noise from the comparative analysis.

[7] We present a comparative analysis between RQ1, RQ2, RQ3, and RQ4. Notice that we create the same bubble charts for comparing strategies, goals, limitations, evaluation metrics, and target users for the data extracted from practice in RQ4. However, due to scarcity of data, we decided to remove the graphics from the manuscript since the bubble charts were almost empty. We provide such bubble charts in our repository: https://zenodo.org/records/10262145

7. Threats to validity and limitations

This section presents a discussion about the internal validity, construct validity, and external validity threats of the presented results. We highlight the limitations and how we tried to mitigate them. In addition, we discuss and point out the limitations that we have not been able to mitigate and should be considered when interpreting the results of our study.

7.1. Internal validity

Internal validity is defined as the extent to which the study design, methodology, and data collection process accurately capture and represent the identified studies. We have identified three internal validity threats in our study: *selection bias, data extraction bias, and inter-rater reliability*.

We acknowledge that selecting relevant studies and MDSE tools for inclusion in our study may introduce a *selection bias*. To mitigate this threat, we have defined a set of exclusion and inclusion criteria so the reviewers' selection is as objective as possible. Moreover, we rely on a research-supported and industry-relevant market study—i.e., GMQ—to select the MDSE tools that we use for answering RQ4. However, the bias of the GMQ and reviewers' subjective judgement in selecting the MDSE tools could result in other relevant studies not being selected.

Data extraction during our study introduces a risk of bias in how the data are collected, introducing a *data extraction bias validity threat*. To mitigate this threat, we ask reviewers to extract as much text as possible that could have relevant information around the specific RQ for which they were extracting data. Regarding the MDSE tool data extraction, we rely on publicly available documentation, and we extracted as much data as possible from such documentation. Moreover, the first and second authors of this paper reviewed the extracted data and supervised that the data other reviewers extracted seemed to be as complete as possible. Nevertheless, human error is inevitable. In addition, we recognise that MDSE tools, especially, have limited documentation on proposals for assistance during modelling tasks. This means that such MDSE tools may implement modelling assistance proposals but not necessarily document them. This causes our extracted data on MDSE tools in the market to be limited to documented proposals to assist users during modelling tasks. Finally, we consider it relevant to mention that our study relies on the authors' terminology from both the literature and the MDSE tool documentation. This means that the data collected could be biased to the authors' terminology, leading to inconsistencies. For example, some authors define their proposal as a tool, but other authors may consider the same proposal as a technique. This shows the need to establish a common terminology in our field to avoid these terminological problems.

We identified a validity threat of subjective interpretation regarding the clusters we proposed in our study. As mentioned earlier, we primarily relied on terminology extracted directly from authors' studies to propose clusters that represent, as closely as possible, the studies grouped within them. However, we noticed that the ontological background for proposing such clusters appears to be subjective and open to various interpretations, potentially introducing bias into our research findings. To address this concern, we implemented rigorous data collection and analysis methods. Moreover, we conducted a triangulation after proposing each cluster. We have measured the level of agreement between the first clustering and after the triangulation. We observe a K-statistic = 0.651, showing a substantial agreement between the first clusters and the final cluster discussion based on Landis and Koch's interpretation of the K-statistic [15] (0.80 > *K*-statistic > 0.61). Nevertheless, it is crucial to recognise that the subjective nature of the ontological background may have influenced the interpretation of results. To mitigate this validity threat, we have reported our clustering process as transparently as possible, making available both data extraction and triangulation information. Moreover, this concern has prompted future work on validating the data we extracted, involving external reviewers and peer review processes.

Involving several reviewers in the selection of studies introduces a threat to inter-rater reliability. We have involved three reviewers in the process and used the K-statistic to measure the effect of the *inter-rater reliability threat* in our results. We observed a K-statistic = 0.634, showing a substantial agreement between reviewers based on Landis and Koch's interpretation of the K-statistic [15] (0.80 > *K*-statistic > 0.61). Having calculated the K-statistic allows us to see that there is an agreement between reviewers for selecting studies. However, we did not measure K-statistic during data extraction since our data were mainly text. Despite this, we consider it important to emphasise that we have made several rounds with the authors of this paper to cluster the extracted data and reach an agreement between us, as mentioned before. In addition, we recognise that this level of agreement may be affected by reviewer fatigue. To this end, the reviewers had enough time to extract the data and read the assigned studies.

7.2. Construct validity

Construct validity refers to the degree to which our study captures and represents the relevant concepts or phenomena of interest using the right tools and tests. We have identified two construct validity threats: *grey literature bias and search bias.*

Initially, we conducted a systematic mapping in search of proposals that assist during modelling. That triggered a *grey literature bias* since our systematic mapping focuses on published literature, excluding relevant information from *grey literature.* Mitigating this threat was a priority since the MDSE field is extremely practical, and not all data is published in scientific literature. Therefore, we decided to run a review of the MDSE tools on the market using the GMQ. This allowed us to analyse grey literature, such as documentation, websites, and user guides. However, we recognise that other grey literature could have been included but was not, such as reports and white papers.

Regarding the *search bias validity threat*, we mitigate this threat by using two different search strategies in our systematic mapping study: database search and forward/backward snowballing. Despite this, we recognise that the selection of scientific databases (in this case, five different ones) can trigger the search bias validity threat. In addition, the selection of the initial studies to perform snowballing introduces a search bias. However, we highlight that we have chosen five relevant databases in software engineering and chose a strategy to select the initial studies for snowballing to mitigate this thread. Regarding the selection of MDSE tools, using a single study triggers the *search bias validity threat.* Nevertheless, we consider that GMQ follows a research-based strategy to gather an overview that is as representative of the MDSE tool market as possible. However, in future work, we consider it relevant to explore other studies or conduct an independent search to find MDSE tools on the market.

7.3. External validity

External validity is concerned with whether we can generalise and apply our research findings beyond the specific context and sample used. We have identified a *language bias* in our study. We include only proposals written in English. Then, studies written in other languages are underrepresented. However, we consider our study a representative sample since most of the literature on software engineering is written in English.

8. Conclusions and future work

In this paper, we have presented results around a main research question: what proposals exist in the literature and practice to assist users during modelling tasks in MDSE tools? To do so, we have conducted a systematic mapping study and explored the state of the practice

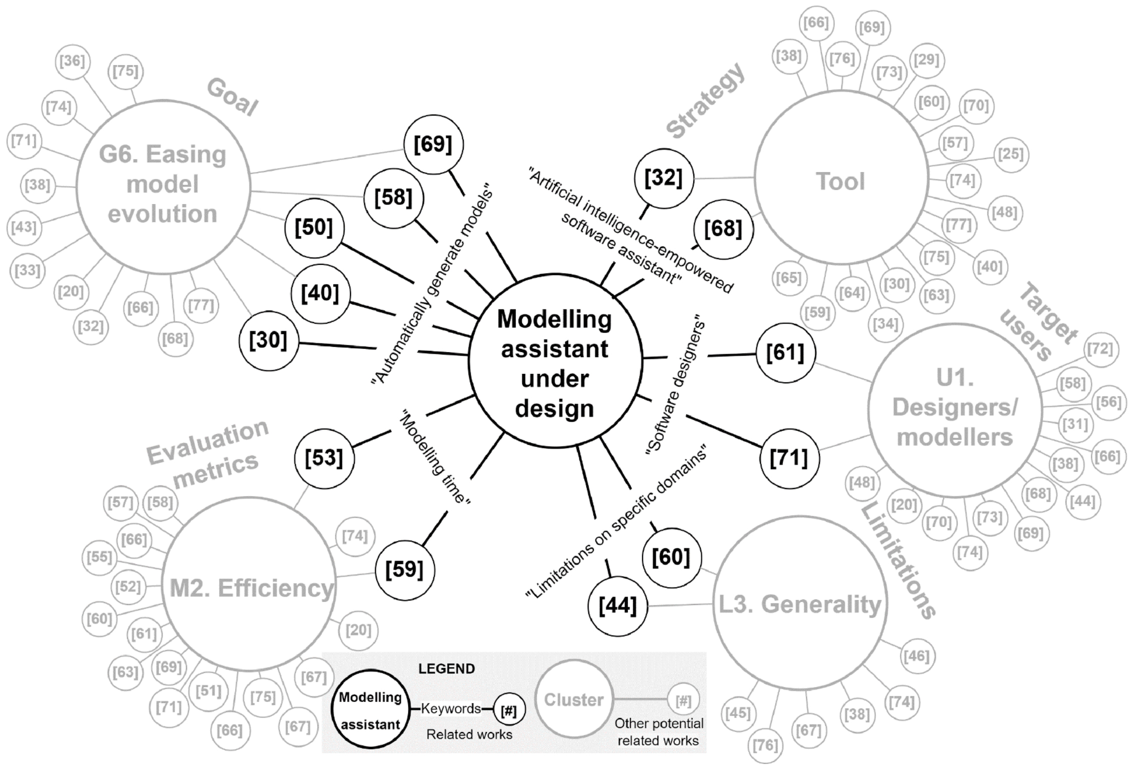

Fig. 14. Example of visualisation in a public repository for designing modelling assistants based on our study clusters.

of assisting users during modelling tasks. As a result, three reviewers selected 58 proposals from a set of 3.176 records to answer the following research questions: i) how is software modelling assisted? ii) what goals and limitations do existing modelling assistance proposals report? and iii) which evaluation metrics and target users do existing modelling assistance proposals consider? As a result, we reported clusters on modelling assistance strategies, goals, limitations, evaluation metrics, and target users that emerged from the extracted data. Moreover, we have reviewed the documentation on 17 MDSE tools available in practice based on the Gartner Magic Quadrant for enterprise low-code application platforms. Using tools documentation, we extracted data to answer iv) what is the state of the practice on modelling assistance?

After analysing the results of our study (see Sections 4, 5, and 6), we have observed that, both in the literature and in practice, there is interest in assisting users during modelling tasks in MDSE tools. Proof of that is that we found 58 studies in the literature specifically aiming to assist users during modelling tasks. On the other hand, we observed answering RQ4 that many MDSE tools we reviewed from the market have no documentation of how they intended to assist users during modelling tasks. However, many of the leading MDSE tools based on GMQ have such documentation, showing that leader MDSE tools are considered relevant to mention how they assist users during modelling tasks. Furthermore, after conducting a comparative analysis, we observed that MDSE tools in practice mentioned common goals, limitations, evaluation metrics, and target users with proposals in the literature. Overall, this shows the relevance of designing in detail how to assist users during modelling tasks in practice and literature.

Despite the observed significance of assisting during modelling tasks, our data from RQ2, RQ3, and RQ4 reveals a gap in both the literature and market-ready proposal. Specifically, we have observed that information regarding their limitations, evaluation metrics, and target users is either scattered or non-existent despite having data on goals and strategies for assisting in modelling tasks. This hinders carrying out an objective comparison between proposals.

In contrast with the results, nowadays, we face disruptive new technologies such as artificial intelligence. These edge technologies represent a disruptive shift in the strategies for assisting users during modelling tasks that we have reported in our study (RQ1) and are awaiting to come in the following years. We based our observation on the introduction of Large Language Models (LLMs) and Generative-Pretrained Transformers (GPTs) to assist diverse users on other software development tasks than modelling, such as generating code based on unstructured natural language [101].

Recognising the potential for disruptive changes in software modelling assistance strategies motivates providing a shared framework where researchers can query the current modelling assistance strategies (RQ1) and the goals pursued through these strategies (RQ2). This stands in contrast to the inherent limitations within each proposal (RQ2). By doing so, new technologies driven by artificial intelligence can leverage existing knowledge, avoiding the reinvention of the wheel and addressing previously identified limitations.

From the non-technical modelling assistance design, we would like to emphasise the significance of explicitly specifying the users whom a proposal intends to assist during modelling (RQ3). Previous experiences—such as the one reported on [102]—emphasise that the success of innovations in MDSE development tools is greatly influenced by the users—a.k.a. user profiles—who interact with these tools. The authors state that lacking a definition of the users hindered their expectation management and user interaction fine-tuning process. This is why the lack of explicit definition in the proposals compiled in our study is of concern to us.

In view of these considerations, we believe that a common framework is indispensable for facilitating the design of novel proposals for software modelling assistance in the years to come. In previous work, some authors have proposed a framework to design modelling assistants [103]—a.k.a. Intelligent Modelling Assistants (IMAs). This framework aims to facilitate modelling assistance design by providing a set of quality aspects the modelling assistants should address—e.g., model quality, semantic quality, confidence, and autonomy, among others. In contrast, other authors propose an emerging framework to elicit

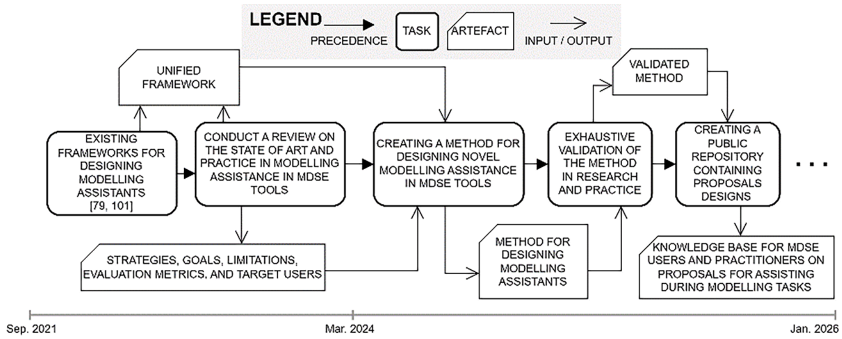

Fig. 15. Research agenda.

requirements for modelling assistants based on target users' input [81]. We believe that, as future work, a significant synergy could be achieved by proposing a unified framework that leverages the concepts already introduced in [81,103] along with the findings regarding strategies, goals, limitations, evaluation metrics, and target users from our study. All these efforts are towards facilitating and improving the designs of future modelling assistants for MDSE tools.

We recognise that more research is needed to fully utilise the results of our study and realise the full potential of a unified framework for designing modelling assistants in MDSE tools. Locally, our focus will be on further design, validation, and method engineering efforts to provide such a framework. However, we believe that a solo effort would not be sufficient to exploit the potential of our results. That is why we plan to create a public repository based on the clusters proposed in this study. The aim of this repository is to connect new modelling assistant designs with related works, identifying similarities and potential synergies. Additionally, it will be a resource for the community to analyse and compare existing proposals. For example, a software development team introduces a novel artificial intelligence-empowered software assistant aiming to automatically generate models, assisting software designers in a specific domain and decreasing the modelling time. With an existing repository that follows the clusters proposed in our study, we could graphically visualise (see Fig. 14) how such a modelling assistant is related to existing works. With the support of the community nurturing this repository with their designs, the modelling assistant connections with existing works can be sustained over time and evolve.

In conclusion, we underscore the vital role of both local-level and community-level efforts in advancing the landscape of modelling assistance. We reflect on our next steps in a research agenda in Fig. 15.

Competing interest statement

The authors declare no competing financial or non-financial interests that could influence the interpretation of the findings or bias the publication of this research.

Funding sources

This research was fully supported by the ZHAW Institute for Applied Information Technology (InIT), the Innosuisse Flagship SHIFT project, and the ZHAW School of Engineering.

CRediT authorship contribution statement

David Mosquera: Writing – original draft, Visualization, Validation, Methodology, Formal analysis, Conceptualization. **Marcela Ruiz:** Writing – original draft, Validation, Supervision, Methodology, Conceptualization. **Oscar Pastor:** Supervision, Conceptualization. **Jürgen Spielberger:** Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

We have shared the data on a public repository and is referenced in the paper.

Acknowledgements

We would like to express our sincere gratitude to Anastassios Martakos and Flavio Einsenring for their invaluable contribution as reviewers in the systematic mapping protocol, and the members of the ZHAW Software Engineering Research Group for participating in the evaluation and conception of the main research questions of our study. Moreover, we would like to extend our sincere gratitude to the anonymous reviewers for their valuable insights, constructive feedback, and meticulous evaluation of our manuscript during the peer review process. Their contributions have greatly enhanced the quality and clarity of this paper.

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